Bachelor-~~thesis~~ arbeit

Tackling Instability in Markowitz's Minimum Volatility Portfolios

Eventuell auch Deutsch, ist aber eine Geschmacksfrage

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Zweitgutachter: ...

Die selbständige und eigenhändige Anfertigung versichere ich an Eides Statt.

Berlin, den August 24, 2021 Francisco Jose Manjon Cabeza Garcia

Zusammenfassung

"contains" oder "suffers from"?

Abstract

After identifying potential investments, asset managers face the question about how to allocate the limited resources they have to each of them, such as to reduce risk while maintaining their target return. This task is of great importance for the asset managers' mandate to generate high risk adjusted returns. Harry M. Markowitz was the first to develop and present a systematic approach to this asset allocation problem in 1952, earning him a Nobel Memorial Prize in Economic Sciences in 1990. However, Markowitz's portfolio construction model presents some pitfalls in practical implementations that limit its performance out-of-sample. In this paper, we analyse Markowitz's Portfolio Construction and its pitfalls. Moreover, we present ~~2~~ two approaches to improve asset allocations by tackling the main critique points of Markowitz's approach. Finally, we compare the results of the optimisation methods on the basis of multi-asset portfolios with the help of ETFs.

Contents

1 Introduction **2**

1.1 Practical Implementation Framework . . . 2

1.1.1 The N Assets . . . 2

1.1.2 Backtest Settings . . . 7

1.1.3 Portfolio Analyses . . . 8

1.1.4 Benchmark Portfolio Construction Methods . . . 9

2 Markowitz's Optimal Portfolio **12**

2.1 N-Assets Optimal Portfolio . . . 12

2.2 Practical Implementation . . . 14

2.2.1 Markowitz's Curse . . . 14

3 Improving Markowitz's Portfolio Model **15**

3.1 Covariance Matrix Shrinking . . . 15

3.1.1 Practical Implementation . . . 15

3.2 Hierarchical Risk Parity . . . 15

3.2.1 Practical Implementation . . . 15

4 Conclusion **16**

4.1 Equal Weighted Portfolios . . . 20

4.2 Inverse Variance Portfolios . . . 20

4.3 Markowitz Minimum Variance Portfolios . . . 20

1 Introduction

How to allocate the limited resources across the set of investable assets available? This is the challenge all investors are faced with, because no investor has unlimited capital available to invest. This task gets more complicated when investors have to take into account their risk appetite and their target return. This asset allocation problem can be defined as an optimisation problem with ~~k~~ two variations:

1. Maximise returns while maintaining risk within a given risk tolerance level, or
2. Minimise risk for a given level of target return.

Referenz

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Harry M. Markowitz was the first to develop and present a systematic approach to this asset allocation problem in both variations in 1952, by defining risk as the standard deviation of past returns and returns as the expected value of past returns. His work earned him a Nobel Memorial Prize in Economic Sciences in 1990. Markowitz's portfolio construction model laid the ground for quantitative asset allocation models. Many asset allocation models today are developed using Markowitz's model and adjusting it to account for modern investors' needs. For example, by using risk measures other than the standard deviation of past returns, or by targeting functions other than expected returns.

The main critique point of Markowitz's portfolio construction model is the lack of stability that leads to a high portfolio turn-over and to concentrated portfolios, specially when correlations rise. In this paper, we analyse Markowitz's portfolio construction model and the causes of low stability by focusing on the minimum volatility portfolio. Moreover, we present some techniques that try to improve the stability of minimum volatility portfolios: 1 regarding parameter estimation for Markowitz's model and 1 alternative allocation method. Moreover, we also compare Markowitz's minimum volatility portfolios to ~~2~~ two naive allocation methods: equal weight and inverse variance allocation.

One?

1.1 Practical Implementation Framework

To show the problems of Markowitz's minimum volatility portfolios in real applications, and compare these portfolios to the portfolios computed using the other methods and techniques, each of the techniques described in this paper is implemented for a portfolio of exchange traded products (ETPs) representing 4 assets classes: equities, fixed income, real estate and commodities. We use QuantConnect ~~a~~ [3] platform for the implementation and as the data source. This platform and its API allow us to perform backtests with one ~~k~~-minute resolution data.

the

1.1.1 The N Assets

Due to the lack of Euro denominated ETP data in the QuantConnect data bank, we decided to use US dollar denominated ETPs which are intended for US based traders. For the selection of ETPs, we use iShares website [26], because it provides a screener of its ETPs[1].

For the equity ETPs, we selected equity index exchange traded funds (ETFs) with regional category "Global" by sectors and industries. From all of the ETPs fulfilling those conditions shown in the online screener, we chose the following ETFs (alphabetically ordered by ticker):

[1] Disclosure: the author of this paper is not associated with iShares, BlackRock or any of its companies.

An dieser Stelle ist unklar, was hiermit gemeint ist. Was ist ein "backtest", wofür wird das gebraucht und ist "one-minute" schnell?

- iShares Global Industrials ETF with ticker "EXI" which seeks to track the investment results of an index composed of global equities in the industrials sector. More specifically, the index tracked is the S&P Global 1200 Industrials Sector Index (TM). The top sub-sectors represented in the ETF holdings are capital goods, transportation and commercial and professional services [10].

- iShares MSCI Global Energy Producers ETF with ticker "FILL" which seeks to track the investment results of an index composed of global equities of companies primarily engaged in the business of energy exploration and production. More specifically, the index tracked is the MSCI ACWI Select Energy Producers Investable Market Index. The top sub-sectors represented in the ETF holdings are integrated oil and gas, oil and gas exploration and production, oil and gas refining and marketing and transportation, coal and consumable fuels and oil and gas drilling [11].

Sind solche Details wichtig? Vielleicht kann hier nur eine Liste mit den Namen stehen und der Rest im Appendix.

- iShares Global Infrastructure ETF with ticker "IGF" which seeks to track the investment results of an index composed of developed market equities in the infrastructure industry. More specifically, the index tracked is the S&P Global Infrastructure Index. The top sub-sectors represented in the ETF holdings are utilities, transportation and energy [13].

- iShares Global Financials ETF with ticker "IXG" which seeks to track the investment results of an index composed of global equities in the financials sector. More specifically, the index tracked is the S&P Global 1200 Financials Sector Index (TM). The top sub-sectors represented in the ETF holdings are banks, diversified financials and insurance [17].

- iShares Global Healthcare ETF with ticker "IXJ" which seeks to track the investment results of an index composed of global equities in the healthcare sector. More specifically, the index tracked is the S&P Global 1200 Healthcare Sector Index (TM). The top sub-sectors represented in the ETF holdings are pharmaceuticals, biotech and life sciences and health care equipment and services [18].

- iShares Global Tech ETF with ticker "IXN" which seeks to track the investment results of an index composed of global equities in the technology sector. More specifically, the index tracked is the S&P Global 1200 Information Technology Sector Index (TM). The top sub-sectors represented in the ETF holdings are software and services, tech hardware and equipment, semiconductors and semiconductor equipment [19].

- iShares Global Comm Services ETF with ticker "IXP" which seeks to track the investment results of an index composed of global equities in the communication services sector. More specifically, the index tracked is the S&P Global 1200 Communication Services 4.5/22.5/45 Capped Index. The top sub-sectors represented in the ETF holdings are interactive media and services, integrated telecommunication services, movies and entertainment, cable and satellite and wireless telecommunication services [20].

- iShares Global Utilities ETF with ticker "JXI" which seeks to track the investment results of an index composed of global equities in the utilities sector. More specificaly, the index tracked is the S&P Global 1200 Utilities (Sector) Capped Index. The top sub-sectors represented in the ETF holdings are electric utilities, multi-utilities, gas utilities and water utilities [21].

- iShares Global Consumer Staples ETF with ticker "KXI" which seeks to track the investment results of an index composed of global equities in the consumer staples sector. More specifically, the index tracked is the S&P Global 1200 Consumer Staples Sector Capped Index. The top sub-sectors represented in the ETF holdings are food, beverages and tobacco, household and personal products and food and staples retailing [22].

- iShares Global Materials ETF with ticker "MXI" which seeks to track the investment results of an index composed of global equities in the materials sector. More specifically, the index tracked is the S&P Global 1200 Materials Sector Index (TM). The top sub-sectors represented in the ETF holdings are chemicals, metals and mining, construction materials and containers and packaging [23].

- iShares Global REIT ETF with ticker "REET" which seeks to track the investment results of an index composed of global real estate equities in developed and emerging markets. More specifically, the index tracked is the FTSE EPRA Nareit Global REITS Net Total Return Index. The top sub-sectors represented in the ETF holdings are retail REIT's, industrial REIT's, residential REIT's and specialised REIT's [24]. This single equity ETF is used as a proxy to represent the real estate asset class.

- iShares Global Consumer Discretionary ETF with ticker "RXI" which seeks to track the investment results of an index composed of global equities in the consumer discretionary sector. More specifically, the index tracked is the S&P Global 1200 Consumer Discretionary Sector Capped Index. The top sub-sectors represented in the ETF holdings are retailing, autos and components, consumer durables and consumer services [25].

Vielleicht Summe $ anstatt U.S. dollar

In figures 1 and 2, we can see the end-of-day absolute return on an investment of 1 U.S. dollar in each of the selected equity ETPs from end 2019 until end 2020. The zero return level and the start of the backtesting period are marked by a horizontal and a vertical black dashed line respectively. The plot to the left of the vertical black dashed line represents the performances in the period used to warm-up the algorithm with data to be able to compute the first portfolio at the beginning of January 2020. The steep drop in both charts around March 2020 marks the results of the COVID-19 crisis.

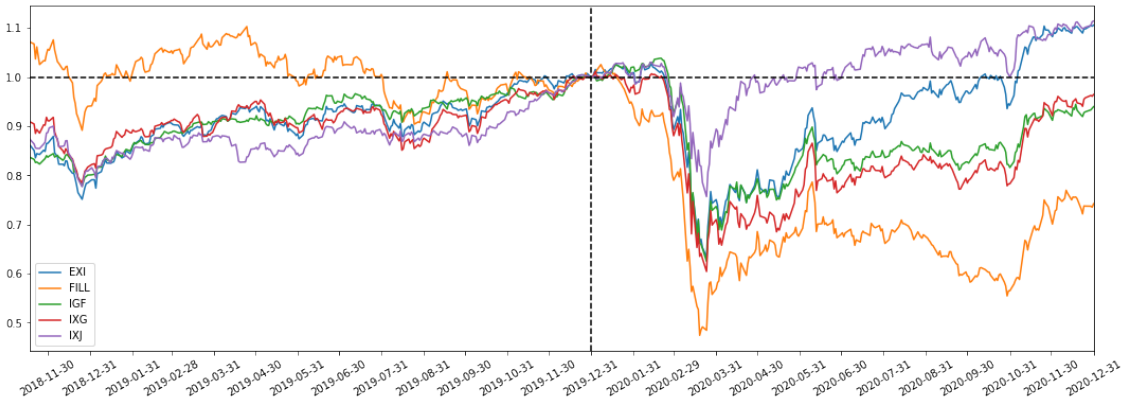

Figure 1: Equity ETPs' returns on a 1 U.S. dollar investment from end 2019 until end 2020

In figure 2, we observe the great run and out-performance of the iShares Global Tech ETF with ticker IXN.

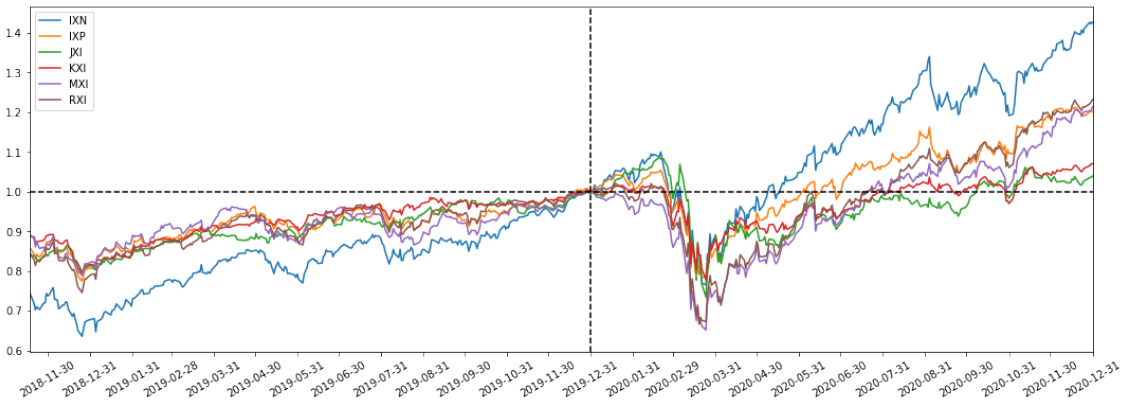

Figure 2: Equity ETPs' returns on a 1 U.S. dollar investment from end 2019 until end 2020

In figure 3, we can see the end-of-day absolute return on an investment of 1 U.S. dollar in the ETP selected as a proxy to represent the real estate asset class.

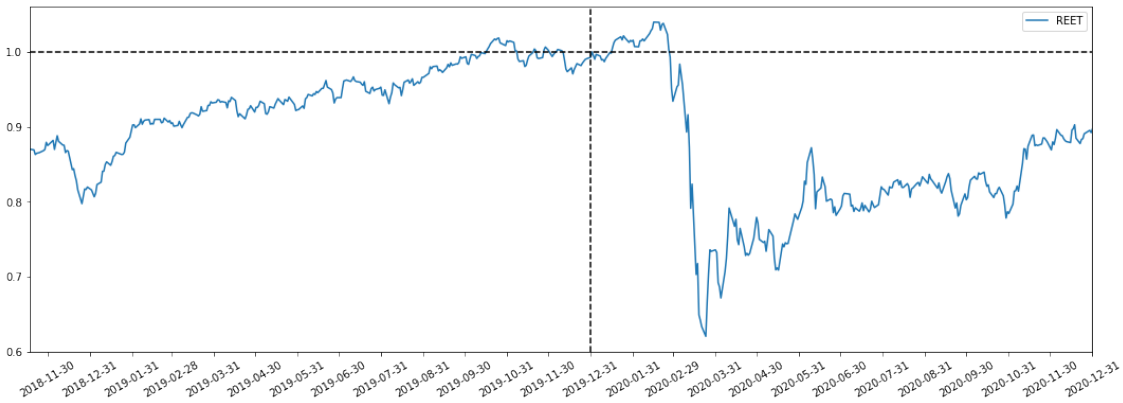

Figure 3: Real Estate ETP returns on a 1 U.S. dollar investment from end 2019 until end 2020

For the fixed income ETPs, we selected fixed income index exchange traded funds (ETFs) of the iShares *Core* category. From all of the ETPs fulfilling those conditions shown in the online screener, we chose the following ETFs (alphabetically ordered by ticker):

- iShares Core International Aggregate Bond ETF with ticker "IAGG" which seeks to track the investment results of an index composed of global non-U.S. dollar-denominated investment-grade bonds that mitigates exposure to fluctuations between the value of the component currencies and the U.S. dollar. More specifically, the index tracked is the Bloomberg Barclays Global Aggregate ex USD 10% Issuer Capped (Hedged) Index. Most of the ETF holdings are rated single A or higher, and have maturities between 3 and 10 years [12].

- iShares Core 10+ Year USD Bond ETF with ticker "ILTB" which seeks to track the investment results of an index composed of U.S. dollar-denominated bonds that are rated either investment-grade or high-yield with remaining maturities greater than 10 years. More specifically, the index tracked is the Bloomberg Barclays U.S. Universal 10+ Year Index. Most of the ETF holdings are rated AAA or BBB [14].

- iShares Core 5-10 Year USD Bond ETF with ticker "IMTB" which seeks to track the investment results of an index composed of U.S. dollar-denominated bonds that are rated either investment-grade or high-yield with remaining effective maturities between 5 and 10 years. More specifically, the index tracked is the Bloomberg US Universal 5-10 Years Index. Most of the ETF holdings are rated AAA [15].

- iShares Core 1-5 Year USD Bond ETF with ticker "ISTB" which seeks to track the investment results of an index composed of U.S. dollar-denominated bonds that are rated either investment-grade or high-yield with remaining maturities between 1 and 5 years. Most of the ETF holdings are rated AAA [16].

In figure 4, we can see the end-of-day absolute return on an investment of 1 U.S. dollar in each of the selected fixed income ETPs from end 2019 until end 2020. The zero return level and the start of the backtesting period are marked by a horizontal and a vertical black dashed line respectively. The plot to the left of the vertical black dashed line represents the performances in the period used to warm-up the algorithm with data to be able to compute the first portfolio at the beginning of January 2020. The steep drop in both charts around March 2020 marks the results of the COVID-19 crisis, from which all fixed income ETFs recovered quickier than equity ETFs.

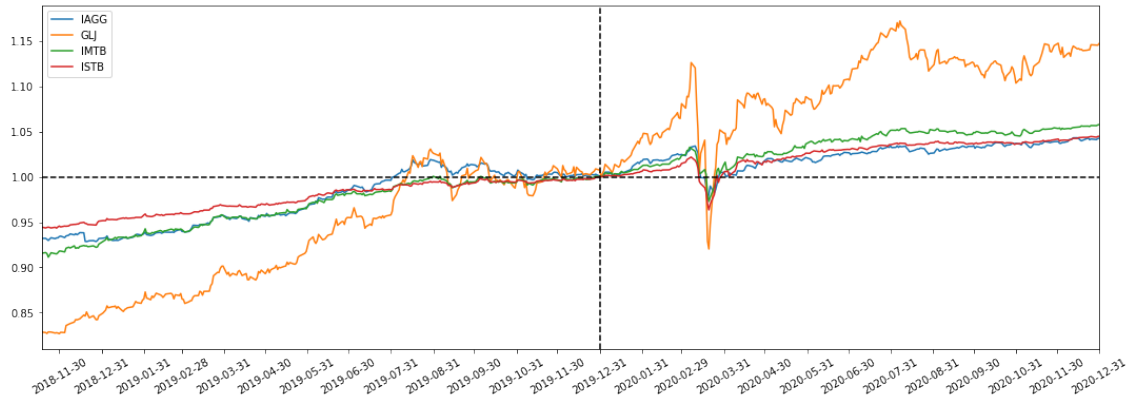

Figure 4: Fixed Income ETPs' returns on a 1 U.S. dollar investment from end 2019 until end 2020

Because iShares commodities ETPs only offer exposure to gold and silver specifically and to certain commodity factor strategies, we turn to Invesco DB Funds for the selection of ETPs that will cover the commodities exposure of our portfolios. We chose Invesco DB Funds because they offer exposure to commodities aggregated by sector[2]. From all commodity-linked funds offered by Invesco, we chose the following ones:

- Invesco DB Agriculture Fund with ticker DBA which seeks to track changes in the level of the DBIQ Diversified Agriculture Index Excess Return over time. The 11 agricultural commodity futures contracts included in the holdings are corn, soybeans, sugar, coffee, live cattle, lean hogs, cocoa, wheat, Kansas wheat, feeder cattle and cotton [6].

- Invesco DB Base Metals Fund with ticker DBB which seeks to track changes in the level of the DBIQ Optimum Yield Industrial Metals Index Excess Return over time. The 3 industrial metals commodity futures contracts included in the holdings are copper, aluminium and zinc [7].

[2]Disclosure: the author of this paper is not associated with any of Invesco's companies.

- Invesco DB Energy Fund with ticker DBE which seeks to track changes in the level of the DBIQ Optimum Yield Energy Index Excess Return over time. The 5 energy commodity futures contracts included in the holdings are WTI crude, gasoline, NY Harbor ULSD, Brent crude and natural gas [8].

- Invesco DB Precious Metals Fund with ticker DBP which seeks to track changes in the level of the DBIQ Optimum Yield Precious Metals Index Excess Return over time. The 2 precious metals commodity futures contracts included in the holdings are gold and silver [9].

Vorher immer klein

In Figure 5, we can see the end-of-day absolute return on an investment of 1 U.S. dollar in each of the selected commodity ETPs from end 2019 until end 2020. The zero return level and the start of the backtesting period are marked by a horizontal and a vertical black dashed line respectively. The plot to the left of the vertical black dashed line represents the performances in the period used to warm-up the algorithm with data to be able to compute the first portfolio at the beginning of January 2020. The steep drop in both charts around March 2020 marks the results of the COVID-19 crisis. The red line represents the returns of the precious metals fund, which recovered quicker than the energy fund in green.

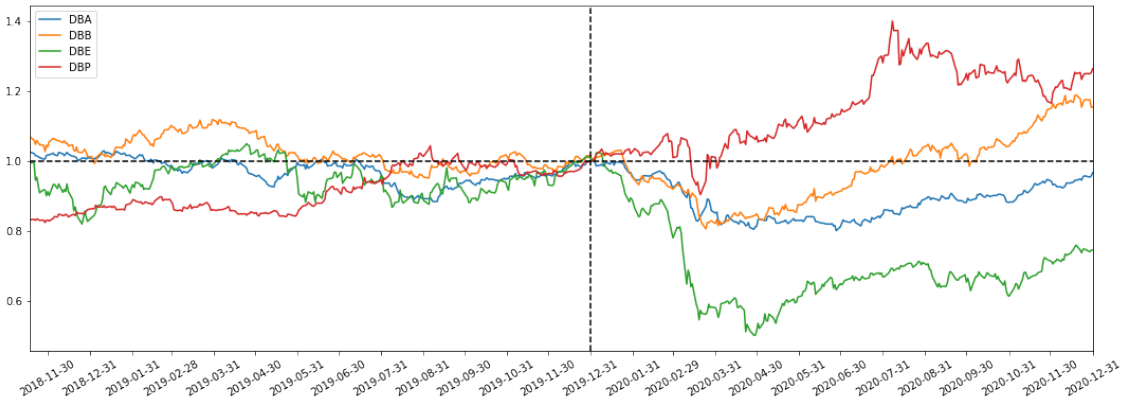

Figure 5: Commodity ETPs' returns on a 1 U.S. dollar investment from end 2019 until end 2020

We acknowledge that the selecting of assets could have been more granular as well as coarser. The ETPs selected might also not be the cheapest available to represent their corresponding index. However, this selection is intended for the illustration of the stability of the different portfolio construction methods presented in this paper only, and not as asset selection for the portfolio construction of a particular investor.

1.1.2 Backtest Settings

In total, 12 portfolios are computed, corresponding to a backtesting period from January 1, 2020 to December 31, 2020 and monthly rebalancing-interval. The rebalancing occurs on the first trading day of each month at 10:00 AM New York time. An initial account balance of 1 million U.S. dollars is assumed, and margin calls are deactivated in order to better simulate the theoretical portfolios computed.

For the computation of portfolio weights, the daily returns of the past 60 days is used as input. This last setting does not affect the equal weight portfolio construction method, because this naive method does not use any past data as input. The reason why we use 60 days of past data is the following. -> siehe nächste Seite

Es wird bisher nicht erklärt, was ein "portfolio" ist und was die "weights" sind.

Let $N \in \mathbb{N} \setminus \{0\}$ be the number of assets considered for the portfolio construction and $T \in \mathbb{N} \setminus \{0\}$ the number of past periods data used as input for the optimisation model. Because the models analysed here rely upon the computation of a covariance matrix (except for the 2 naive methods: equal weight and inverse variance allocation), we need to compute one. Let the matrix that contains the T returns in each row for all N assets be defined as $R \in \mathbb{R}^{N \times T}$. The, $rank(R) \leq \min\{N, T\}$ holds.

Let now $\Sigma := [\sigma_{i,j}]_{i,j=1,\dots,N} \in \mathbb{R}^{N \times N}$ be the covariance matrix of the N assets, where $\sigma_{i,j}$ represents the covariance of the ith and jth asset, then Σ can be computed as

$$\Sigma = \frac{1}{T}(\tilde{R}^T \cdot \tilde{R})$$

where $\tilde{R} \in \mathbb{R}^{N \times T}$ is defined as the matrix R after centring the data row-wise, i.e. after subtracting from each element in each row the expected value of that same row. From this definition of Σ, we get that

Hieraus folgt aber noch nicht $rank(\Sigma) = N$.

$$rank(\Sigma) \leq rank(\tilde{R}) \leq rank(R) \leq \min\{N, T\}$$

That means that $\Sigma \in GL_N(\mathbb{R})$ if only if $rank(\Sigma) = rank(\tilde{R}) = rank(R) = N$, i.e. $N \leq T$ must hold.

In our practical implementation framework, we have selected $N = 20$ assets, therefore $T \geq 20$ must hold. We chose $T = 60$ because it corresponds to 3 full trading months of data.

Ist das der Grund (siehe vorherige Seite)?

1.1.3 Portfolio Analyses

The portfolios generated by each optimisation method in backtesting will be analysed with respect to risk-adjusted return, concentration and stability. The Sharpe ratio is the chosen measure to quantify risk-returns. To assess concentration, we will use a normalised version of the Herfindahl-Hirsch-Index (HHI). The normalisation step is necessary, because we will allow for short positions in our portfolios, and the HHI is only defined for positive portfolio weights. We will measure stability via the turnover of the portfolio.

Definition 1.1 (Sharpe Ratio [5]). *Let $\mu_P \in \mathbb{R}$ be the return of investment P and $\sigma_P \in \mathbb{R}_{\geq 0}$ its standard deviation. Moreover, let $R_f \in \mathbb{R}$ be the risk-free rate. Then the* ***Sharpe Ratio of investment*** *P $SR_P \in \mathbb{R}$ is given by*

Sind "investment P" und "portfolio P" die gleichen Begriffe?

$$SR_P := \frac{\mu_P - R_f}{\sigma_P}$$

Dies ist kein "paper"

Remark 1.2. *In this paper, we do not compute the Sharpe ratio of the backtest separately, and rely upon QuantConnect's internal overall portfolio statistics for its computation, i.e. all Sharpe ratios presented in this paper are the Sharpe ratios that appear in QuantConnect's backtesting results. QuantConnect computes the Sharpe ratio using the same definition as above, but using the annualised return of the portfolio in the backtest [2].*

Definition 1.3 (Herfindahl-Hirsch-Index [1]). *Let P be a portfolio containing N assets with weights $w_i \in \mathbb{R}_{\geq 0}$ for all $i = 1, \dots, N$, such that $\sum_{i=1}^{N} w_i = 1$. We define the* ***Herfindahl-Hirsch-Index (HHI)*** *as ~~the sum of the squares of the weights, i.e.~~*

$$HHI_P := \sum_{i=1}^{N} w_i^2 \quad \dots$$

~~**Definition 1.4** (Normalised Herfindahl-Hirsch-Index). *Let P be a portfolio containing N assets with weights $w_i \in \mathbb{R}$ for all $i = 1, \dots, N$, such that $\sum_{i=1}^{N} w_i = 1$. We define*~~ ... and *the* ***normalised Herfindahl-Hirsch-Index (nHHI)*** *as* ~~*the sum of the squares of the normalised weights, i.e.*~~

$$nHHI_P := \sum_{i=1}^{N} \left(\frac{|w_i|}{\sum_{j=1}^{N} |w_j|} \right)^2.$$

~~**Remark 1.5.** *Let P be a portfolio containing N assets with weights $w_i \in \mathbb{R}$ for all $i = 1, \dots, N$, such that $\sum_{i=1}^{N} w_i = 1$. Because $\sum_{i=1}^{N} |w_i| \geq \left| \sum_{i=1}^{N} w_i \right|$, the following holds*~~

Obviously,

$$\begin{aligned} nHHI_P &= \sum_{i=1}^{N} \left(\frac{|w_i|}{\sum_{j=1}^{N} |w_j|} \right)^2 \\ &= \left(\frac{1}{\sum_{j=1}^{N} |w_j|} \right)^2 \sum_{i=1}^{N} w_i^2 \\ &= \left(\frac{1}{\sum_{j=1}^{N} |w_j|} \right)^2 HHI_P \leq HHI_P \end{aligned}$$

~~*i.e., for a given portfolio P, the Herfindahl-Hirsch-Index is smaller or equal than the normalised Herfindahl-Hirsch-Index.*~~

Definition 1.6 (Turnover [33]). *Consider a portfolio P with N assets. Let $w_{j,t+1} \in \mathbb{R}$ be the portfolio weight assigned to asset i at time t by the portfolio optimisation method, and let $w_{j,t^+} \in \mathbb{R}$ be the portfolio weight of asset i before rebalancing at time $t+1$ for $i = 1, \dots, N$ and $t = 0, \dots, T$. The turnover of the portfolio P is defined as*

$$TO_P := \frac{1}{T} \sum_{t=0}^{T} \sum_{i=1}^{N} \left(|w_{i,t+1} - w_{i,t^+}| \right)$$

where $T \in \mathbb{N}$ is the number of rebalancing points of the portfolio.

This turnover quantity can be interpreted as the average percentage of wealth traded at each rebalancing point.

→ nicht mehr Teil der Definition, sondern "Interpretation"

1.1.4 Benchmark Portfolio Construction Methods

We use 2 naive portfolio allocation models, equal weight and inverse variance allocation, as benchmarks for the other 3 methods (Markowitz minimum volatility portfolio, Markowitz minimum volatility portfolio with covariance matrix shrinking and hierarchical risk parity) to estimate if the the increased complexity of the latter can be justified in form of better risk-adjusted returns, concentration and stability.

Definition 1.7 (Equal Weight Portfolio [33]). *Consider a portfolio of $N \in \mathbb{N}_{>0}$ assets, and let $w \in \mathbb{R}^N$ be the vector of allocation weights. The* ***equal weight allocation portfolio (EWP)*** *is defined as*

$$w_i := \frac{1}{N} \quad \text{for all} \quad i = 1, \dots, N$$

The equal weight portfolio is the most widely used allocation method when no information about the assets in the portfolio is available to rank them or compute some measure to define some other portfolio. It is also the most naive way of allocating capital across the N assets considered. However, this naive method has proven to perform even better than other, even very sophisticated, methods [33].

Definition 1.8 (Inverse Variance Portfolio [33]). *Consider a portfolio of $N \in \mathbb{N}_{>0}$ assets, and let $w \in \mathbb{R}^N$ be the vector of allocation weights. The* ***inverse variance allocation portfolio (IVP)*** *is defined as*

$$w_i := \frac{\frac{1}{\mathbb{V}[X_i]}}{\sum_{k=1}^{N} \mathbb{V}[X_k]} \quad \textit{for} \quad i = 1, \dots, N$$

where $\mathbb{V}[X_i]$ is the variance of the returns' distribution of the i-th asset.

Remark 1.9. *The inverse variance allocation portfolio is the solution of the following optimisation problem*

$$\begin{aligned} \arg\min_{w \in \mathbb{R}^N} \quad & w^T \cdot \Sigma \cdot w \\ s.t. \quad & \mathbf{1}^T \cdot w = 1 \end{aligned}$$

where $\Sigma := diag(\mathbb{V}[X_1], \dots, \mathbb{V}[X_N])$ such that X_i is the returns distribution of the i-th asset.

This means that the inverse variance portfolio is the minimum volatility portfolio considering all assets uncorrelated.

nicht Teil der Definition

Although IVP is more complex than the equal weight allocation model because the estimation of some parameters is needed, it is still considered a naive method.

The vectors with the optimal portfolios computed these 2 naive methods can be found in Appendix. In figure 6, a visual representation of the target allocation of all 20 assets in the IVP over the backtesting period is presented. Most of the portfolio is allocated to ISTB (iShares Core 1-5 Year USD Bond ETF) and IAGG (iShares Core International Aggregate Bond ETF). The allocation to the first drops at the start of the COVID-19 crisis, while the allocation of the second rises. The allocation to IMTB (iShares 5-10 Year USD Bond ETF) rises after the start of the COVID-19 crisis. The rest of assets main a relatively stable allocation over the whole backtesting period. Finally, the high allocation to fixed income ETFs is not surprising, because they tend to present lower variance of returns, while the allocation to equity, including real-estate, and commodity ETPs remains low due to the higher variance of returns that they tend to present.

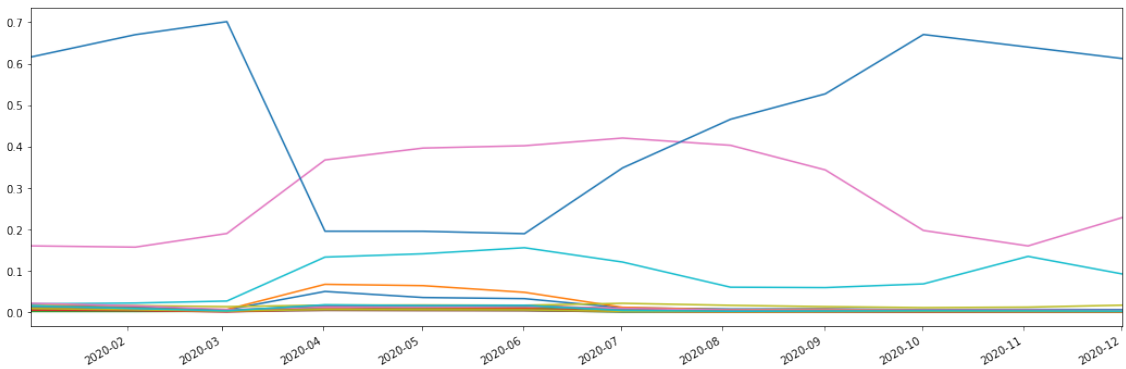

Figure 6: IVP Target Allocation over Time

A similar plot for the EWP allocation over time is not necessary, since the allocation to each asset is equal and remains the same over the whole backtesting period.

Figure 7 shows the evolution of the nHHI of the 2 naive portfolio allocation models described above. The nHHI of the equal weight portfolios is low and stays the same over

time, as expected. In the case of IVP, the nHHI rises with the start of the COVID-19 crisis to latter more than halve, mostly because of a reduction in the allocation to ISTB as explained previously.

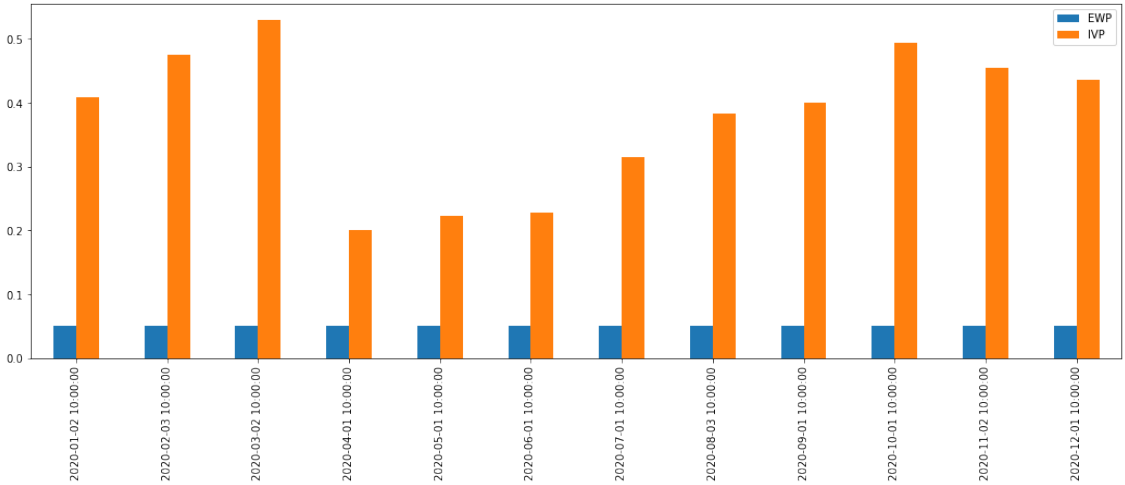

Figure 7: nHHI over Time

Table 1 presents the risk-adjusted returns and turnover of both naive strategies. Inverse variance portfolios achieve a Sharpe ratio of 1.218, which is significantly higher than the Sharpe ratio of equal weighted portfolios.

add turnovers

Portfolio Optimisation Method	**Sharpe Ratio**	**Turnover**
EWP	0.43	not yet computed
IVP	1.218	not yet computed

Table 1: Sharpe Ratio and Turnover Figures

2 Markowitz's Optimal Portfolio

2.1 N-Assets Optimal Portfolio

Here I will give a mathematical description of the optimisation problem presented by Markowitz, including some short proofs of the key points used to formulate the optimal solution as the solution of a linear equation system.

Definition 2.1 (Optimal Mean-Variance Portfolio [28]). *Let $N \in \mathbb{N}_{\geq 0}$ be the number of assets considered for the portfolio construction. Then the vector with the allocation weights for each of the N assets in the* ***optimal portfolio for a given level of return*** *$\mu_P \in \mathbb{R}$ is given by the solution of the following optimisation problem*

$$\begin{aligned} \arg\min_{w \in \mathbb{R}^N} \quad & w^T \cdot \Sigma \cdot w \\ s.t. \quad & \mathbf{1}^T \cdot w = 1 \\ & \mu^T \cdot w = \mu_P \end{aligned}$$

where $\mathbf{1} := [1, \ldots, 1]^T \in \mathbb{R}^N$, $\Sigma \in \mathbb{R}^{N \times N}$ is the covariance matrix of returns and $\mu \in \mathbb{R}^N$ is the vector of expected returns of the N assets.

The set of pairs $(\mu_p, \sigma_P) \in \mathbb{R} \times \mathbb{R}_{\geq 0}$ such that, for each $\mu_P \in \mathbb{R}$, $\sigma_P := w_{opt}^T \cdot \Sigma \cdot w_{opt}$ where w_{opt} is the solution to the optimisation problem above is called the ***efficient frontier****.*

The main critique point of this optimisation problem is the high sensitivity to small changes in the expected returns vector, which leads to significantly different portfolio allocations even for a unchanged covariance matrix [29]. Because returns are difficult to forecast accurately enough to keep the sensitivity within certain limits, some asset managers abstain from using them as input parameter. Portfolio allocations are then computed on the basis of the covariance matrix only, which is a risk parameter, hence the name *risk-based* asset allocation. In the case of Markowitz's portfolio optimisation model, risk-based optimal portfolios correspond to minimum volatility portfolios.

Definition 2.2 (Minimum Volatility Portfolio [32]). *Let $N \in \mathbb{N}_{\geq 0}$ be the number of assets considered for the portfolio construction. Then the vector with the allocation weights for each of the N assets in the* ***minimum volatility portfolio*** *is given by the solution of the following optimisation problem*

$$\begin{aligned} \arg\min_{w \in \mathbb{R}^N} \quad & w^T \cdot \Sigma \cdot w \\ s.t. \quad & \mathbf{1}^T \cdot w = 1 \end{aligned}$$

Theorem 2.3. *Let the covariance matrix $\Sigma \in \mathbb{R}^{N \times N}$ be given. Assuming that Σ is non-singular, the minimum volatility portfolio allocation vector is given by*

$$w_{opt} := \frac{1}{\mathbf{1}^T \cdot \Sigma^{-1} \cdot \mathbf{1}} \cdot \Sigma^{-1} \mathbf{1}$$

Proof. Let $\Sigma \in \mathbb{R}^{N \times N}$ be a non-singular covariance matrix. We define the Lagrangian corresponding to the minimum volatility portfolio optimisation problem as follows

$$\mathcal{L}(w, \lambda) := w^T \cdot \Sigma \cdot w - \lambda(\mathbf{1}^T \cdot w - 1)$$

To find the solution to the optimisation problem, we take the derivatives with respect to both w and λ, ~~and set them to 0~~

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial w} &= 2 \cdot \Sigma \cdot w - \lambda \cdot \mathbf{1}, \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= 1 - \mathbf{1}^T \cdot w. \end{aligned}$$

Setting these to zero gives

$0 = 2\Sigma w - \lambda 1 \iff w = \frac{\lambda}{2}\Sigma^{-1} 1,$

and

$0 = 1 - 1^T w \iff \lambda = \frac{2}{1^T \Sigma 1},$

and hence $w_{opt} = \ldots$.

siehe oben

$$\Rightarrow \frac{\partial \mathcal{L}}{\partial w} = 0 \Leftrightarrow \quad 2 \cdot \Sigma \cdot w = \lambda \cdot \mathbf{1}$$
$$\Leftrightarrow \quad w = \Sigma^{-1} \cdot \left(\frac{\lambda}{2} \cdot \mathbf{1}\right)$$
$$\Leftrightarrow \quad w = \frac{\lambda}{2} \cdot \Sigma^{-1} \cdot \mathbf{1}$$

better form alignment arrows

$$\Rightarrow \frac{\partial \mathcal{L}}{\partial \lambda} = 0 \Leftrightarrow \quad 1 - \mathbf{1}^T \cdot w = 0$$
$$\Leftrightarrow \quad 1 - \mathbf{1}^T \cdot \frac{\lambda}{2} \cdot \Sigma^{-1} \cdot \mathbf{1} = 0$$
$$\Leftrightarrow \quad 1 - \frac{\lambda}{2} \cdot \mathbf{1}^T \cdot \Sigma^{-1} \cdot \mathbf{1} = 0$$
$$\Leftrightarrow \quad \frac{\lambda}{2} \cdot \mathbf{1}^T \cdot \Sigma^{-1} \cdot \mathbf{1} = 1$$
$$\Leftrightarrow \quad \frac{\lambda}{2} = \frac{1}{\mathbf{1}^T \cdot \Sigma^{-1} \cdot \mathbf{1}}$$
$$\Leftrightarrow \quad \lambda = \frac{2}{\mathbf{1}^T \cdot \Sigma^{-1} \cdot \mathbf{1}}$$

better form alignment arrows

$$w_{opt} := \frac{\frac{2}{\mathbf{1}^T \cdot \Sigma^{-1} \cdot \mathbf{1}}}{2} \cdot \Sigma^{-1} \cdot \mathbf{1} = \frac{1}{\mathbf{1}^T \cdot \Sigma^{-1} \cdot \mathbf{1}} \cdot \Sigma^{-1} \mathbf{1}$$

□

expression for w_{opt} ~~relies on computing~~ contains the inverse of the covariance matrix ~~used as input~~. Because covariance matrices are only positive-semidefinite, i.e. not necessarily non-singular, and because computing the inverse of a matrix is computationally very expensive, we ~~would like to redefine the analytical solution~~ compute w_{opt} as the solution of a linear system. ~~However, we will further assume that the covariance matrix used is non-singular~~.

as follows:

Theorem 2.4. *~~Let the covariance matrix $\Sigma \in \mathbb{R}^{N \times N}$ be given. Assuming that Σ is non-singular,~~ the ~~minimum~~ volatility portfolio allocation ~~vector~~ can ~~be computed~~ as follows:*

1. *Solve the linear system $\Sigma \cdot w = \mathbf{1}$, and define its solution as $\tilde{w}_{opt} \in \mathbb{R}^N$. (Note that if Σ is non-singular, $\tilde{w}_{opt}$ might not be unique or even exist).*

   Hier könnte man diskutieren, wie man w_{opt} im singulären Fall bestimmt, z.B. als "least squares solution."

2. *Normalise $\tilde{w}_{opt}$ for its elements to sum up to 1, i.e. set $w_{opt} = \frac{1}{\mathbf{1}^T \cdot \tilde{w}_{opt}} \cdot \tilde{w}_{opt}$.*

Proof. Let ~~the covariance~~ matrix $\Sigma \in \mathbb{R}^{N \times N}$ ~~be~~ given ~~and non-singular. Moreover,~~ let $\tilde{w}_{opt}$ ~~be the solution of the linear system $\Sigma \cdot w = \mathbf{1}$. Then we can write $\tilde{w}_{opt} = \Sigma^{-1}\mathbf{1}$~~.

$$\cancel{\Rightarrow \frac{1}{\mathbf{1}^T \cdot \tilde{w}_{opt}} = \frac{1}{\mathbf{1}^T \cdot \Sigma^{-1} \cdot \mathbf{1}} \cdot \Sigma^{-1}\mathbf{1} = w_{opt}}$$

□

Dies ist kein "Theorem", sondern eigentlich "offensichtlich".

2.2 Practical Implementation

For the practical implementation, we use Numpy's linear system solver, which computes a solution to the linear system in Theorem 2.4 using LAPACK routine _gesv. This means, that the solution is computed using the LU decomposition of the covariance matrix with partial pivoting.

Theorem 2.5 (LU-Decomposition with Partial Pivoting [30]). *Let $A \in \mathbb{R}^{N \times N}$ be a non-singular matrix, then there exists a unique factorisation*

$$A := P \cdot L \cdot U$$

where $P \in \mathbb{R}^{N \times N}$ is a permutation matrix, $L \in \mathbb{R}^{N \times N}$ is a unit lower triangular matrix and $U \in \mathbb{R}^{N \times N}$ is an upper triangular matrix.

Figure 8 shows the evolution of allocations in Markowitz minimum volatility portfolios over time. The changes in allocation are clearly more pronounced than in the case of IVPs (see figure 6), specially around the start of the COVID-19 crisis when correlations rose and all asset classes suffered negative returns.

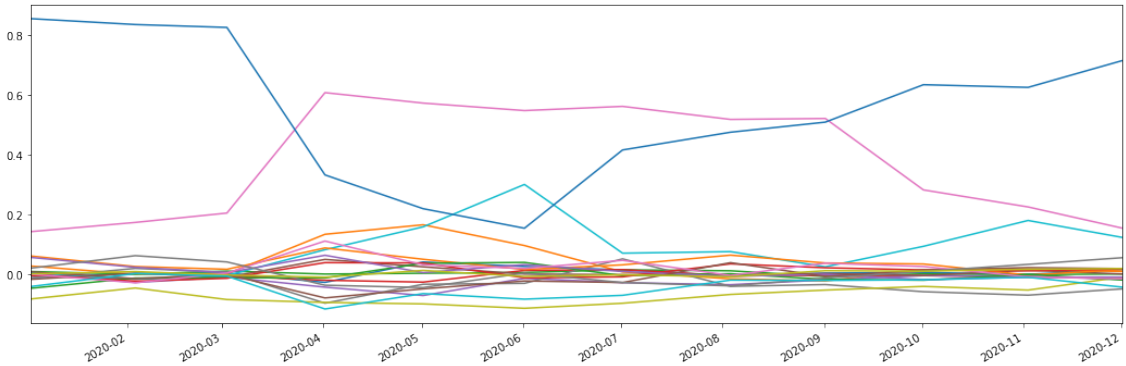

Figure 8: MMVP Target Allocation over Time

The concentration of MMVPs measured by the nHHI remains however lower than in IVPs over the whole backtesting period as figure 9 shows. Nevertheless, a drop in concentration can be observed following the start of the COVID-19 crisis.

With respect to risk-adjusted returns, MMVPs deliver significantly worse results than both naive allocation methods. The Sharpe ratio of Markowitz's strategy is even negative because MMVPs fail to generate positive returns during the backtesting period.

nover fig-

Portfolio Optimisation Method	Sharpe Ratio	Turnover
EWP	0.43	not yet computed
IVP	1.218	not yet computed
MMVP	-0.023	not yet computed

Table 2: Sharpe Ratio and Turnover Figures

2.2.1 Markowitz's Curse

The main reason for the lack of stability is what Marcos López de Prado calls *Markowitz's curse*: the more correlated the investments, the greater the need for diversification, and yet the more likely we will receive unstable solutions because the condition number of the correlation matrix rises, which leads to high numerical errors.

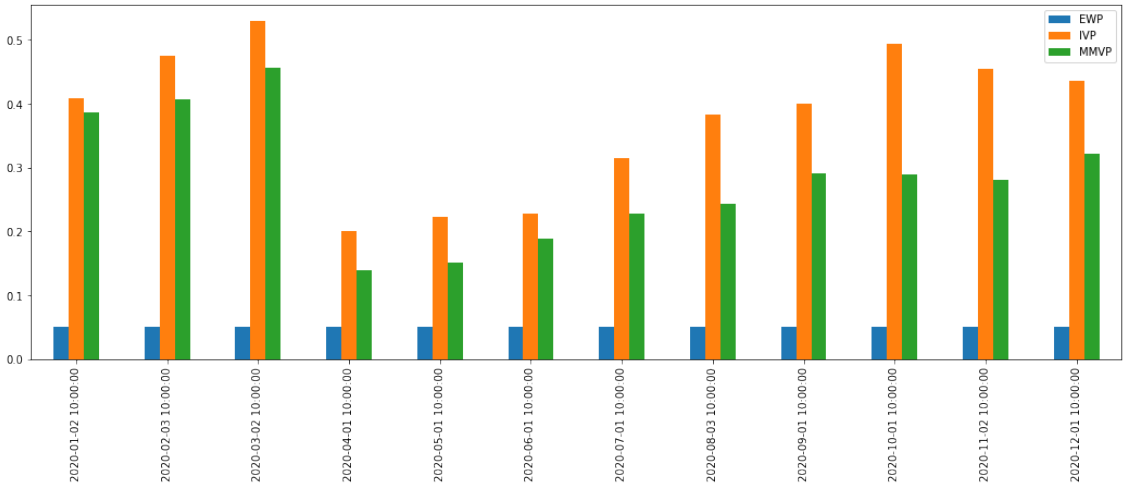

Figure 9: nHHIn over Time

3 Improving Markowitz's Portfolio Model

In this section, I will present and analyse 4 different approached to improve the stability of optimal portfolios. I will keep the description of the first 3 methods short. However, I don't want to remove them completely because I would like to compare the results of those 3 "simpler" methods to the "more complex" Hierarchical Risk Parity method.

3.1 Covariance Matrix Shrinking

This method tries to improve stability by reducing noise in the sample covariance matrix. It is easy to implement and I believe it is also to interpret by analysing the condition number of the shrunk covariance matrix.

References: [31]

3.1.1 Practical Implementation

Analyse results comparing the method to Markowitz's method.

3.2 Hierarchical Risk Parity

This subsection, together with subsection 2.1, is the biggest part of the thesis. I will describe the method in a more mathematical way, as we already discussed. Moreover, I will discuss the differences between this method and the original method of Markowitz, the model assumptions and also its limitations.

References: [4], [32], [27]

3.2.1 Practical Implementation

Analyse results comparing the method to Markowitz's method and also to the other 3 methods above.

4 Conclusion

To conclude, I will present which method performed better and the consequences of having more stable portfolios: less rebalancing necessary, lower transaction costs, etc. I would also like to shortly discuss what the alternative methods can still not achieve and some ideas to further improve López de Prado's method. However, I will keep this very last part and won't get into details of how to solve the problems or how to precisely improve López de Prado's algorithm.

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Appendix

4.1 Equal Weighted Portfolios

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;DBA;DBB;DBE;DBP;EXI;FILL;IAGG;IGF;ILTB;IMTB;ISTB;IXG;IXJ;IXN;IXP;JXI;KXI;MXI;REET;I
2020-01-02 10:00:00;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;
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2020-12-01 10:00:00;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;0.05;

4.2 Inverse Variance Portfolios

ormat

;DBA;DBB;DBE;DBP;EXI;FILL;IAGG;IGF;ILTB;IMTB;ISTB;IXG;IXJ;IXN;IXP;JXI;KXI;MXI;REET;I
2020-01-02 10:00:00;0.008213;0.009619;0.002294;0.012815;0.013;0.005073;0.160481;0.016445;0.020122;0.020
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2020-04-01 10:00:00;0.050581;0.067742;0.015817;0.015726;0.013423;0.004738;0.367811;0.007286;0.017612;0
2020-05-01 10:00:00;0.035906;0.064526;0.005898;0.016118;0.012562;0.004254;0.396495;0.006766;0.017907;0
2020-06-01 10:00:00;0.033183;0.048473;0.005104;0.016197;0.012378;0.003996;0.402141;0.006841;0.017337;0
2020-07-01 10:00:00;0.011031;0.012317;0.001094;0.00901;0.003427;0.00175;0.42069;0.004201;0.021895;0.121
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2020-10-01 10:00:00;0.006605;0.004829;0.001652;0.001592;0.002748;0.001418;0.197724;0.003206;0.0115;0.06
2020-11-02 10:00:00;0.007275;0.004512;0.001582;0.001947;0.002616;0.00148;0.160375;0.003414;0.012732;0.1
2020-12-01 10:00:00;0.006758;0.003763;0.001234;0.003508;0.002321;0.00097;0.229325;0.003013;0.017463;0.0

4.3 Markowitz Minimum Variance Portfolios

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;DBA;DBB;DBE;DBP;EXI;FILL;IAGG;IGF;ILTB;IMTB;ISTB;IXG;IXJ;IXN;IXP;JXI;KXI;MXI;REET;I
2020-01-02 10:00:00;-0.010414;0.028871;0.010633;-0.003647;0.004386;-0.015516;0.14375;-0.011566;-0.08113
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2020-03-02 10:00:00;0.005609;0.002881;0.01212;-0.004221;-0.008074;-0.008607;0.205867;0.008143;-0.082986
2020-04-01 10:00:00;-0.025722;0.134976;0.002177;-0.018601;-0.041597;0.051082;0.608069;-0.095099;-0.0925
2020-05-01 10:00:00;0.042397;0.166745;0.004891;-0.024824;-0.069568;0.026318;0.573183;-0.031942;-0.09755
2020-06-01 10:00:00;0.027575;0.097344;0.00755;0.014299;-0.012156;0.002142;0.548158;-0.028851;-0.11225;0
2020-07-01 10:00:00;0.016022;0.00925;0.016267;0.016942;-0.026539;0.000338;0.562201;0.052863;-0.095358;0
2020-08-03 10:00:00;-0.004167;-0.013779;0.013066;-0.003591;-0.033396;0.000732;0.518622;-0.038851;-0.0659
2020-09-01 10:00:00;0.00538;-0.020803;-0.005947;-0.001729;-0.017461;0.003833;0.521779;-0.032946;-0.05142
2020-10-01 10:00:00;0.007559;0.016467;0.002709;-0.002947;-0.003141;-0.002827;0.28321;-0.056866;-0.03909
2020-11-02 10:00:00;0.001158;0.018739;0.00029;-0.007918;-0.00491;0.01496;0.226281;-0.068394;-0.051375;0.
2020-12-01 10:00:00;0.016859;0.010583;0.002467;-0.009781;-0.015243;0.002338;0.155022;-0.047214;-0.00772